NTNU
Kunnskap for en bedre verden

Department of Engineering Cybernetics

TTK4551 - Engineering Cybernetics, Specialization Project

Comparison of Skyfield SGP4 and Basilisk Numerical Orbit Propagation for GNSS-R Formation-Flying Applications

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17th December 2025

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CHAPTER

ONE

NOTES

Strongest narative for my thesis:

1. My specialization project showed that single-TLE SGP4 is not sufficient as ground truth, and that simplified drag limited the conclusions.
2. The Master's thesis improves upon the validation reference by using regularly update TLEs, making Skyfield-SGP4 a short-horizon, observation-updated reference
3. Basilisk is then validated as sufficiently accurate for long time-horizon mission simulation
4. The validated Basilisk propagator is used inside a realistic two-satellite GNSS-R formation simulator
5. Mission lifetime is estimated from fuel depletion + formation loss, with battery/power constraints shaping operations.

Scope: Focus on:

- Basilisk propagation validation
- Two-satellite deployment and formation acquisition
- Autonomous mode switching
- battery and fuel lifetime curves
- formation evolution
- mission lifetime estimate

Include, but keep on a high level:

- GNSS-R signal physics (just what is required to enable observation)
- Advanced controller derivations (both attitude and formation)

CHAPTER

TWO

INTRODUCTION

https://www.itk.ntnu.no/emner/masteroppgave

This has to be included in the report to make it clear to the sensors what has been done:

- Information, software and equipment available/used
- What help has been received during the work
- The use of AI. The AI-declaration schema MUST be inserted on the first page after the title page

2.0.1 FNs sustainability goals

2.0.2 Background and motivation

2.1 Take-aways from the specialization project

2.1.1 Research problem

2.1.2 Research objectives

2.1.3 Research questions

CHAPTER

THREE

LITERATURE REVIEW

The literature review conducted for this paper can be divided into three themes:

- GNSS-R Small-Satellite Missions
- Formation Flying and Formation Maintenance for Small Satellites
- Integrated Satellite Mission Simulation and Lifetime Analysis

3.1 GNSS-R Small-Satellite Missions

- Scientific evolution:
  1. GNSS as a signal of opportunity
  2. GNSS-R feasibility for different remote sensing applications. Bistatic radar principle
  3. Early demonstrations of GNSS-R remote sensing on aircraft mission
  4. Spacecraft demonstrations
  5. Validation of technology for different applications
  6. Expansion of GNSS-R Mission applications (include jamming/spoofing detection)
- Mapping useful GNSS-R missions after validation of technology at spacecraft altitudes
- Feasibility of cooperative swarms and their respective formation types w. benefits

Reflected Global Navigation Satellite Systems (GNSS) signals were first discussed as a

3.2 Formation Flying and Formation Maintenance for Small Satellites

3.3 Integrated Satellite Mission Simulation and Lifetime Analysis

3.4 Research gaps and thesis contributions

CHAPTER

FOUR

BACKGROUND THEORY

4.1 Reference Frames

A reference frame $\mathcal{F}^{rf} : \left\{\mathcal{O}_{rf};\ \hat{\mathbf{x}}_{rf}, \hat{\mathbf{y}}_{rf}, \hat{\mathbf{z}}_{rf}\right\}$ is uniquely defined by three dextral orthogonal unit vectors $\{\hat{\mathbf{x}}_{rf}, \hat{\mathbf{y}}_{rf}, \hat{\mathbf{z}}_{rf}\}$ and the origin $\mathcal{O}_{rf}$ [1]. This section will present mathematical definitions for the most important reference frames used in the thesis. The individual reference frame indices are assigned to be consistent with the conventional Basilisk notation.

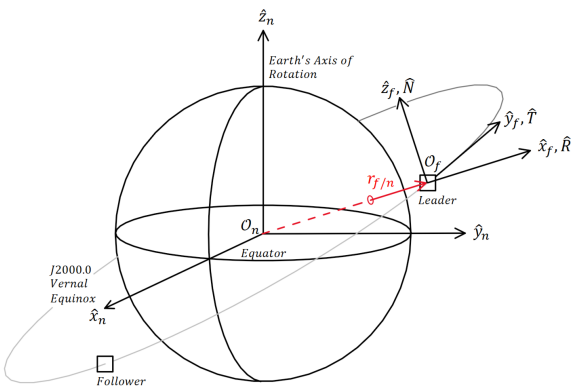

Figure 4.1.1: Illustration of the ECI ($\mathcal{F}^n$)and RTN ($\mathcal{F}^f$) reference frame definitions. The basis unit vectors are displayed with different lengths for illustrative purposes. Adopted from [2, 3] and modified from [4]

.

4.1.1 J2000 Earth-Centered Inertial (ECI) Frame

The ECI reference frame, $\mathcal{F}^{rf} : \{\mathcal{O}_n;\ \hat{\mathbf{x}}_n, \hat{\mathbf{y}}_n, \hat{\mathbf{z}}_n\}$, is fixed relative to the distant stars, and is therefore commonly approximated as an inertial frame in which Newtonian mechanics are valid [5]. The frame is defined with its origin in the Earth's Center of Mass (CM), with $\hat{\mathbf{z}}_n$ pointing along its rotational axis towards the geographical North Pole, $\hat{\mathbf{x}}_n$ pointing towards the vernal equinox through the equator, and $\hat{\mathbf{y}}_n$ defined according to the right-hand rule. Multiple variants of the ECI definition exists, primarily varying in the epoch used to define the Earth's equatorial plane and equinox direction. In this thesis, the J2000 ECI frame is adopted. It is defined using the J2000.0 reference epoch, equivalent to 1st January 2000, at 12:00 Terrestrial Time (TT) [6]. The frame definition is illustrated in Figure 4.1.1.

4.1.2 Radial-Transverse-Normal (RTN) Frame

The Radial-Transverse-Normal (RTN) reference frame, $\mathcal{F}^{f} : \left\{\mathcal{O}_f;\ \hat{\mathbf{x}}_f, \hat{\mathbf{y}}_f, \hat{\mathbf{z}}_f\right\}$, is a local orbital frame that is useful for expressing relative motion between spaceborne objects. Vallado presents the frame as the *Satellite coordinate system, RSW* in [7], but the *RTN* name is chosen for this thesis to better convey each component's physical significance. For use in formation flight, the coordinate frame origin is assigned to the leader satellite, l's CM. The dextral orthogonal unit vectors are defined with the $\hat{\mathbf{x}}_f$ pointing in the radial direction along the satellite's Earth-relative position vector, the $\hat{\mathbf{z}}_f$ axis points normal to the orbital plane spanned by the position and velocity vector, and $\hat{\mathbf{y}}_f$ fulfills the right-hand rule. Note that the latter will always point in direction of positive velocity, but will only be perpendicular to the velocity for circular orbits [7, 8]. Using this definition, the frame can be expressed mathematically as:

$$\hat{\mathbf{R}}^n = \hat{\mathbf{x}}_f^n = \frac{\mathbf{r}_{l/n}^n}{||\mathbf{r}_{l/n}^n||}, \qquad \hat{\mathbf{N}}^n = \hat{\mathbf{z}}_f^n = \frac{\mathbf{r}_{l/n}^n \times \mathbf{v}_{l/n}^n}{||\mathbf{r}_{l/n}^n \times \mathbf{v}_{l/n}^n||}, \qquad \hat{\mathbf{T}}^n = \hat{\mathbf{y}}_f^n = \hat{\mathbf{z}}_f^n \times \hat{\mathbf{x}}_f^n \tag{4.1}$$

Here, $r_{l/n}^n$ and $v_{l/n}^n$ are used to denote the Earth-relative leader position and velocity expressed in ECI, while $\hat{\mathbf{y}}_f$ and $\hat{\mathbf{z}}_f$ correspond to the along- and cross-track directions, respectively. An illustration of the leader satellite's position relative to $\mathcal{O}_n$ and its affixed RTN frame is shown in Figure 4.1.1.

4.1.3 Body-Fixed Frame

The body-fixed reference frame, $\mathcal{F}^{b} : \{\mathcal{O}_b;\ \hat{\mathbf{x}}_b, \hat{\mathbf{y}}_b, \hat{\mathbf{z}}_b\}$, is a moving reference frame that is rigidly attached to the spacecraft body. Because of this property, the body-fixed frame is used for expressing satellite attitude, actuator torques and sensor measurements, since these quantities are naturally expressed relative to the body itself. The orientation of its axes are typically chosen to best represent the spacecraft geometry, and to coincide with the principal axes of inertia [9]. Consequentially, CubeSat body axes are commonly selected to be perpendicular to its side panels. The frame origin is commonly placed at the CM in order to simplify the rotational equations of motion, and to reduce translational-

rotational coupling [10]. An illustration of the body frame definition for a 1U CubeSat is shown in Figure 4.1.2.

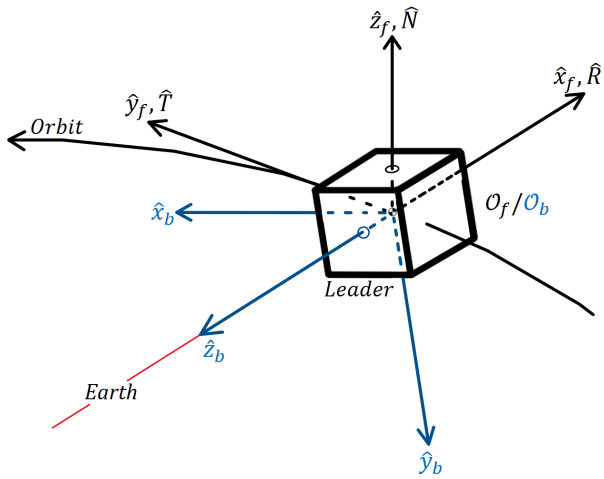

Figure 4.1.2: Illustration of the body-fixed ($\mathcal{F}^b$)and RTN ($\mathcal{F}^f$) reference frame definitions for a 1U CubeSat. Adopted from [11].

4.1.4 Desired Attitude Frame

$\mathcal{F}^r : \{\mathcal{O}_f;\ \hat{\mathbf{x}}_r, \hat{\mathbf{y}}_r, \hat{\mathbf{z}}_r\}$

4.2 Modified Rodrigues parameters

Modified Rodrigues Parameters (MRP)s are 3-element attitude parameterization commonly used for spacecraft attitude control. The representation can be derived from the Euler-Rodrigues symmetric parameters, better known as the unit quaternion:

$$\mathbf{q} = \begin{bmatrix} \boldsymbol{\epsilon} \\ \eta \end{bmatrix} = \begin{bmatrix} \epsilon_1 & \epsilon_2 & \epsilon_3 & \eta \end{bmatrix}^T, \qquad \mathbf{q}^T\mathbf{q} = |\boldsymbol{\epsilon}|^2 + \eta^2 = 1 \tag{4.2}$$

where η is the scalar component and $\boldsymbol{\epsilon}$ is the imaginary vector. Using this definition, Shuster writes in [12] that the MRPs are obtained through a stereographic projection of the unit quaternion onto $\mathbb{R}^3$

$$\mathbf{p} = \frac{\boldsymbol{\epsilon}}{1+\eta}, \qquad \mathbf{m} = \frac{\boldsymbol{\epsilon}}{1-\eta} \tag{4.3}$$

Here, $\mathbf{p}$ and $\mathbf{m}$ are the positive and negative forms, differing only in the singularity placement. This thesis utilizes the positive form, and will hereby be referred to as $\boldsymbol{\sigma}$. Using the axis-angle representation of a rotation, where $\hat{\mathbf{n}}$ represents the unit rotation axis and θ is the angle of rotation about the axis, the positive-form MRPs can be expressed

as:

$$\mathbf{q} = \begin{bmatrix} \hat{\mathbf{n}} \sin(\theta/2) \\ \cos(\theta/2) \end{bmatrix} \quad \Rightarrow \quad \boldsymbol{\sigma}(\theta, \hat{\mathbf{n}}) = \hat{\mathbf{n}} \tan\left(\frac{\theta}{4}\right), \tag{4.4}$$

From this formulation, it follows that the MRPs become singular as $\theta \to 2\pi$, since

$$\tan\left(\frac{\theta}{4}\right) \to \infty \quad \text{for} \quad \theta \to 2\pi \tag{4.5}$$

To avoid numerical issues near the singularity, an equivalent MRP shadow set, $\boldsymbol{\sigma}^s$, may be used [13]. The shadow set transformation is defined as:

$$\boldsymbol{\sigma}^s = -\frac{\boldsymbol{\sigma}}{\boldsymbol{\sigma}^T \boldsymbol{\sigma}} \tag{4.6}$$

Switching to the shadow set when $||\boldsymbol{\sigma}|| > 1$ effectively bounds the MRP set by the unit sphere, and prevents the parameterization from approaching the singularity numerically [14, 13, 12].

4.3 Classic Orbital elements

The classic orbital elements are a set of six parameters used to uniquely define the shape, size and orientation of an orbital plane, as well as the position of an orbiting body. In astrodynamics, these parameters provide a compact and intuitive representation of orbital motion, commonly used for orbital analyses and spacecraft guidance. The classic orbital elements used in this thesis are:

a :	semimajor axis	orbital plane size and shape
e :	eccentricity	
i :	inclination	orbital plane orientation
Ω :	right ascension of the ascending node (RAAN)	
ω :	argument of perigee	
M :	spacecraft	orbital body position

It is important to note that the semimajor axis, and mean anomaly are in some orbital element configurations replaced with the angular momentum h, and the true anomaly θ, respectively. In any case, these six parameters define a Keplerian orbit, and they can be grouped according to the orbital property they describe: The semimajor axis and eccentricity define the shape and size of the orbital plane, while the inclination, RAAN and argument of perigee describe the plane's orientation. Finally, the mean anomaly is used to describe the spaceborne object's position along the orbital trajectory. An illustration of the classic orbital elements is shown in Figure 4.3.1.

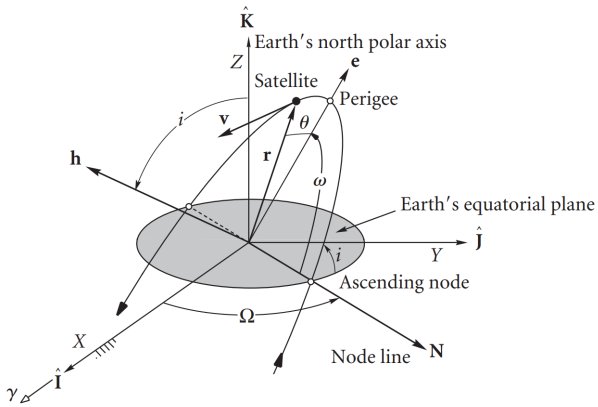

Figure 4.3.1: Orbital elements relative to the ECI frame. Courtesy of [15]

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The semimajor axis is defined as half of the longest ellipse diameter, and determines the overall size of the orbit. For elliptical orbits, this parameter is directly coupled with the orbital period through Kepler's third law. The eccentricity is a measure of how much the orbital trajectory deviates from a circle. It takes values in the span between zero and one inclusive, where an eccentricity of zero corresponds to a perfectly circular orbit, and an increasing eccentricity correspond to elliptical orbits of increasing elongation.

The inclination is the intersection angle between the orbital plane and the reference plane, commonly the equatorial plane spanned by the $\hat{x}_n$ and $\hat{y}_n$ axes in ECI. This angle determines the tilt of the orbital plane, while its orientation about the $\hat{z}_n$ axis is determined by the RAAN parameter. For nonzero inclinations, the orbital trajectory intersects the reference plane at two points, the ascending and descending nodes. The ascending node is where the object travels from the Southern to the Northern Hemisphere, and RAAN is defined as the angle between this node and the $\hat{x}_n$ axis in the equatorial plane. The argument of perigee specifies the orientation of the ellipse within the orbital plane. It is defined as the angle from the ascending node to the perigee along the orbital trajectory, where perigee is a term specific to Earth satellites describing the point along the trajectory closest to Earth. Finally, the mean anomaly defines the orbiting body's position along the orbit as a function of time. It is defined as the elapsed fraction of the orbital period since the orbiting body passed the perigee [15, 16].

4.4 Orbital perturbations

4.4.1 Spherical harmonics

4.4.2 Atmospheric drag

4.4.3 NRLMSISE-00 (MSIS) Atmosphere model

4.4.4 Solar radiation pressure

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4.7.2 Cold gas thrusters

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4.8 GNSS-R operations

4.8.1 Pointing requirements

4.8.2 Formation constraints

4.9 Simplified General Perturbations 4 (SGP4)

CHAPTER

FIVE

METHODOLOGY

- Should include a mass budget table similar to 'Table 4-4: SSTL-Cube Mass Budget Breakdown per Sub-System' in the HYDROSWARM paper

CHAPTER

SIX

RESULTS

Plotting ideas for part 2:

- The big important plot: Fuel level over time
- Each subsystem should be validated in addition to showing the full fuel consumption over simulation duration:
  - RW speed/torque step response
  - Thruster force output as a function of fuel level (/tank pressure)
  - MRP feedback controller tracking error and/or step response (would also be interesting to see a step response for different fuel levels)
  - Decomposed net power and battery level (should also show the illumination factor in the same/adjacent plot and the FSW state)

CHAPTER

SEVEN

DISCUSSION

IMPORTANT!

I have done some initial testing using the Basilisk Station Keeping example, and it seems like a strict CPO or Concentric formation is unachievable for the spacecraft reconfig module. If I have time, I should absolutely implement a HCW/ROE based controller for formation control. If I don't have time, I should just argue that this should be expanded in the future, and make a solid discussion about how my fuel consumption estimate is in comparison to the more realistic HCW case.

CHAPTER

EIGHT

CONCLUSIONS

8.1 Conclusion

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APPENDICES

A - GITHUB REPOSITORY

All simulator code and LaTeX-files used in this document are included in the GitHub repositories linked below.

Specialization project report LaTeX repository link

- https://github.com/chystad/Specialization-Project-Report-LaTeX-

Orbit Simulator repository link

- https://github.com/chystad/Orbit_Simulator

B - NUMERICAL VALUES

Numerical Values for Physical Constants

$Constant$	$Value$	$Unit$
J_0	-1	-
J_1	0	-
J_2	1.08263×10^{-3}	-
J_3	-2.53266×10^{-6}	-
J_4	-1.61962×10^{-6}	-
P_{Sun}	$4.578299965 \times 10^{-6}$	Ws/m^3
$1AU$	149597871	km
R	6378136.6	m
H_0	7200	m
ρ_0	1.225	kg/m^3
$-$	$-$	$-$
$-$	$-$	$-$
$-$	$-$	$-$

Table .0.1: Physical constants and associated numerical value. J-terms courtesy of